

CENTRALIZED SALES AI AGENT PLATFORM

Unified Research, Architecture & Technical Design

Integrated design for six connected AI sales agents

Platform scope	Specialized agents
Shared intelligence	Knowledge-Based Agent
Research & qualification	Lead Research Agent
Campaign creation	Campaign Content Agent
Execution	Outreach Agent
Conversation intelligence	Inbox / Reply Intelligence Agent
Decision layer	Lead Lifecycle / Next Action Agent

Design objective

Create one governed sales system where each specialist agent has a clear job, uses shared context, returns structured results, and never acts outside the permissions and approval rules of the platform.

Prepared from the six supplied agent architecture and research documents. The report consolidates their common patterns into one platform design.

1. Executive Summary

Why the six agents should be designed as one platform

The supplied designs describe six specialist agents that solve different parts of the sales process. The common architectural pattern is a centralized orchestrator, a shared knowledge/context layer, persistent lead state, tool-based execution, event-driven updates, and human approval for risky actions.

- Lead Research finds and verifies useful person, company and buying-signal information before outreach.
- Campaign Content turns a structured brief into strategy, content, checks, approvals and performance learning.
- Outreach coordinates prospecting, qualification, personalization, execution, follow-up and CRM updates.
- Inbox / Reply Intelligence searches live mail, understands conversations, prioritizes replies and drafts responses.
- Lead Lifecycle / Next Action keeps one current stage per lead and chooses the next best action.
- Knowledge-Based Agent is the shared intelligence layer that retrieves evidence and assembles task-specific context for every other agent.

The central idea

The agents should not operate as six independent applications. They should work as specialist services around one sales state. The orchestrator routes work, the knowledge layer provides trusted context, the lifecycle agent decides what should happen next, action agents execute approved tasks, and every result becomes a new event.

Simple mental model

KNOW → RESEARCH → QUALIFY → CREATE → ACT → UNDERSTAND REPLY → DECIDE NEXT → UPDATE KNOWLEDGE

Expected benefits

Area	Platform outcome
Consistency	All agents use the same lead, account, campaign and conversation context.
Quality	Important claims carry evidence, source and confidence.
Automation	Routine research, content, outreach and inbox work can run with fewer manual steps.
Control	Suppression, permissions, approval gates and audit logs are enforced centrally.
Learning	Sends, replies, meetings, corrections and outcomes update shared state and knowledge.

2. Platform Scope and Agent Responsibilities

Six agents, six clear responsibilities

Agent	Primary responsibility	Main output
Knowledge-Based	Collect, normalize, store, retrieve and assemble trusted context.	ContextPackage
Lead Research	Identify people/companies, enrich data, find signals, score ICP fit and verify evidence.	Lead Research Package
Campaign Content	Plan campaigns, generate channel assets, check brand/claims/compliance and learn from results.	Campaign + approved content assets
Outreach	Discover/enrich/qualify prospects, personalize, execute approved outreach and manage follow-ups.	Outreach action + sequence state
Inbox / Reply	Search/read email, classify intent, summarize threads, draft replies and escalate meaningful conversations.	Reply intelligence + draft/handoff
Lead Lifecycle	Maintain stage, interpret events, choose next action, enforce guardrails and route approval.	NextActionDecision

Shared responsibilities

- Every agent receives a task-specific context package instead of raw database access.
- Every agent uses typed tools; the language model proposes tool calls but the backend executes them.
- Every important result is written back as an event or state update.
- Every customer-facing or irreversible action passes through policy and permission checks.
- Every important decision can be traced to evidence, confidence, rule version and actor.

What should not happen

Anti-pattern	Why it is unsafe / difficult
One giant agent doing everything	Large context, unclear ownership, hard testing and weak governance.
Direct LLM access to CRM or mailbox	Model can bypass permission and validation boundaries.
Independent memory per agent	Agents can disagree about lead state and repeat work.
Unverified personalization	Creates hallucinated claims and poor customer experience.
Automatic sending without a gate	High-impact external action is not controlled by policy.
Static knowledge only	The system becomes stale as leads, replies and campaigns change.

3. Main Unified Architecture

The complete platform view

CENTRALIZED SALES AI AGENT PLATFORM

One orchestrator • six specialist agents • shared knowledge and state • governed actions

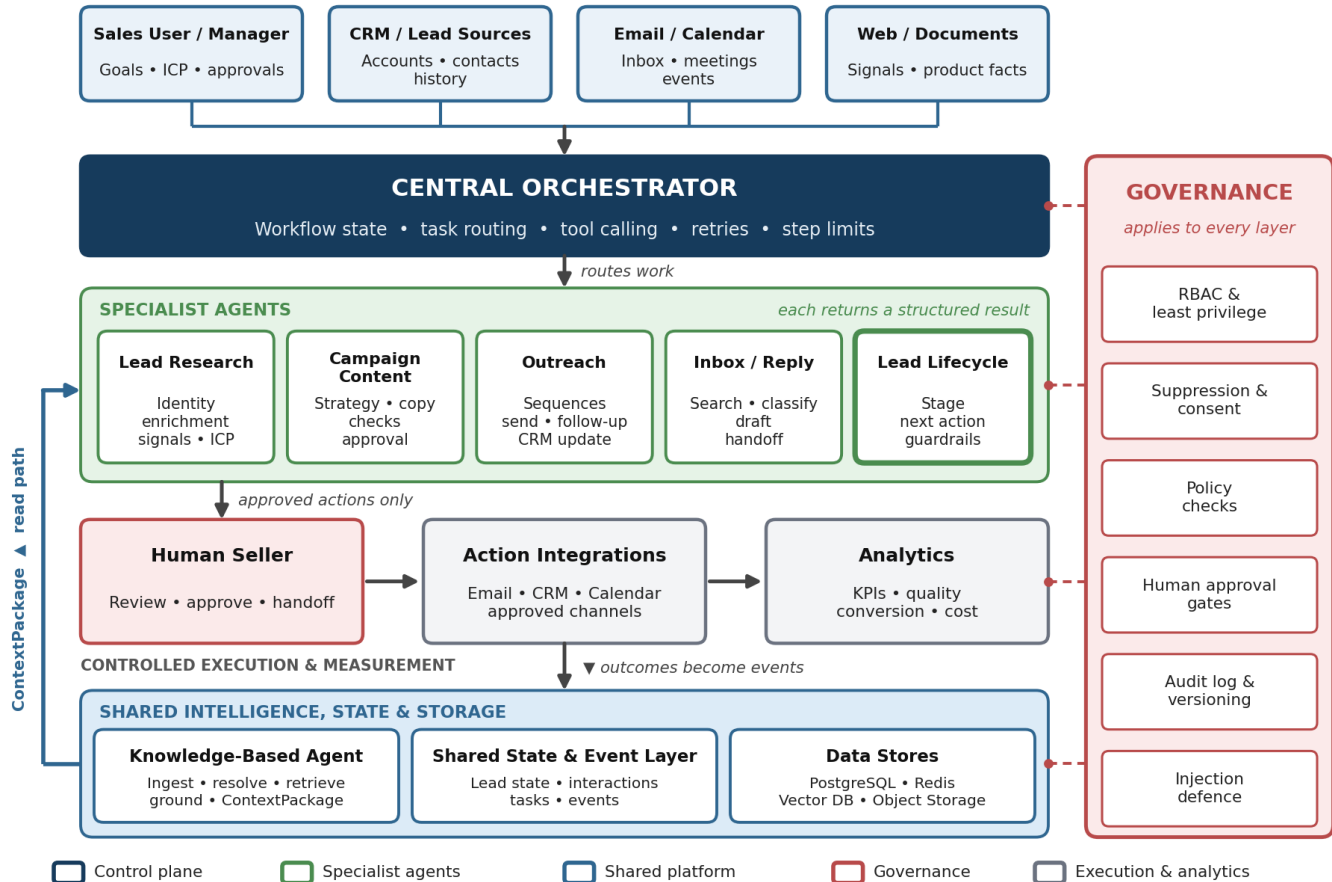


Figure 1. Main architecture of the centralized Sales AI Agent Platform.

The architecture combines the six supplied designs into one control plane. Inputs enter through business systems and user goals. The Central Orchestrator decides which specialist should work. The Knowledge-Based Agent and shared state layer provide common context. Governance applies across the platform. Actions are executed only through approved integrations, and outcomes return as events.

Most important connection

Lead Lifecycle is the decision layer, while Knowledge-Based is the shared context layer. The other specialist agents perform focused work and return structured evidence or outcomes.

4. Shared Knowledge and Context Layer

How all agents use the same trusted information

The Knowledge-Based design treats knowledge as more than a vector database. It combines structured business data, semantic knowledge, event history and current interaction context. It ingests information, resolves entities, stores it, retrieves task-specific evidence and assembles a compact ContextPackage.

Layer	Role	Examples
Sources	Original business and interaction inputs	CRM, product docs, playbooks, emails, calls, research, events
Ingestion	Bring information into the platform	Connectors, APIs, event consumers, file processing
Processing	Make information usable	Parsing, chunking, classification, entity resolution, deduplication, metadata
Stores	Keep different forms of knowledge	PostgreSQL, vector store, event store, object storage
Hybrid retrieval	Find the right context	Keyword + semantic + structured + history + permission filters
Context assembly	Prepare task-specific context	Identity, current state, history, evidence, constraints, conflicts
Governance	Control use of knowledge	RBAC, provenance, confidence, audit, human approval

Why shared context matters

- Lead Research can ask what is already known and what is missing.
- Campaign Content can retrieve persona, product facts, approved messaging and recent interactions.
- Outreach can check lead state, previous sends, channel constraints and timing before execution.
- Inbox can interpret a reply using conversation history, objections and campaign context.
- Lead Lifecycle can combine lifecycle, interactions, engagement, research, open tasks and playbook rules.

The result should be a ContextPackage with identity, current state, relevant knowledge, evidence, constraints and conflict information. This prevents each agent from inventing its own view of the customer.

5. Lead Research Agent

From raw lead to evidence-backed research package

The Lead Research design focuses on accurate identity, current person and company information, business signals, ICP fit, evidence and confidence. It is not a paragraph generator; its output is intended to be safely consumed by downstream agents.

1. Receive lead input and campaign/ICP rules.
2. Match the person and company and check existing CRM data.
3. Research person, company, contact and business signals.
4. Attach source, date, status and confidence to important facts.
5. Calculate ICP fit, intent, timing and confidence using fixed scoring rules.
6. Run suppression and safety checks.
7. Create the Lead Research Package and pass it to the next agent.

Research area	Examples
Person	Name, title, department, seniority, location, work history
Company	Industry, size, location, funding, leadership, hiring, technology, recent activity
Buying signals	Funding, hiring, leadership change, expansion, product launch, technology change, major partnership
Evidence	Claim, source, date, confidence, status
Scoring	ICP fit 40%, intent 30%, timing 20%, confidence 10% as an example framework
Output	Lead identity, company data, signals, evidence, score, compliance and personalization inputs

Waterfall enrichment

The supplied design recommends using the cheapest and most trusted available information first: internal CRM data, then primary enrichment, fallback enrichment when necessary, contact verification, signal research and finally the packaged result. This reduces cost and unnecessary external calls.

6. Campaign Content Agent

Turning a campaign brief into approved multi-channel content

The Campaign Content design treats content creation as a controlled workflow rather than a one-shot generation task. The agent works from a structured brief, retrieves approved brand and product knowledge, generates assets, checks them, routes them for approval, publishes when permitted and learns from performance.

Stage	What happens
Brief	Capture objective, audience, offer, channels, dates, budget and constraints.
Plan	Retrieve brand kit, product facts and past performance; propose message pillars and calendar.
Generate	Create channel-specific posts, emails, ad copy, outlines, scripts and image prompts.
Check	Run brand, claim, compliance, similarity and platform-format checks.
Review	Human edits, comments, approves or rejects; feedback is applied to assets.
Publish	Schedule or publish approved content through integrations.
Measure & learn	Read performance, compare variants and inform the next plan.

Grounding rules

- Brand voice and do-not-say rules are retrieved instead of remembered by the model.
- Product facts, prices, offers and legal disclaimers are treated as approved claims.
- Each important claim should be traceable to the supporting knowledge entry.
- Unsupported claims are flagged rather than silently accepted.
- Human approval is required before publishing or spending budget.

Connection to the other agents

Campaign Content can use Lead Research evidence for audience-specific messaging, Knowledge-Based context for product and brand facts, Outreach state for campaign sequencing, Inbox signals for objections and Lead Lifecycle decisions for when new content is needed.

7. Outreach Agent

The execution engine for qualified prospects

The Outreach design combines prospect discovery, enrichment, research, qualification, personalization, policy checks, execution, reply monitoring, follow-up, CRM memory and analytics. Its core goal is relevant conversations, not simply more messages.

1. Define campaign goal, target audience, offer and constraints.
2. Discover prospects from permitted sources or CRM segments.
3. Enrich missing contact and company information.
4. Research current and relevant signals.
5. Qualify against ICP, intent/readiness and exclusion rules.
6. Prioritize high-fit and high-intent prospects.
7. Personalize from verified evidence and approved value propositions.
8. Validate factual grounding, missing variables, tone and policy.
9. Obtain approval when required and execute the approved action.
10. Track sends, timestamps, content versions and statuses.
11. Monitor replies, bounces and out-of-office events.
12. Classify replies and stop, follow up or hand off as appropriate.
13. Update CRM/shared state and measure performance.

Personalization rule

Use a specific, verified observation connected to a relevant business problem. If a required variable is missing, use a safe generic structure rather than inventing information.

Connection to the platform

Outreach receives research and context from shared knowledge, uses Lifecycle decisions to determine when and how to act, sends through permissioned integrations, and sends every outcome back as an event so Inbox and Lifecycle can react.

8. Inbox / Reply Intelligence Agent

Understanding live conversations and protecting the handoff to humans

The Inbox Agent connects to Gmail or Outlook through OAuth. It searches and reads mailbox data, summarizes threads, classifies intent, identifies messages needing action, drafts replies and can take approved actions. The model never touches the mailbox directly; backend tools execute all operations.

Capability	Platform behavior
Search	Find by sender, date, label, keyword or semantic meaning.
Understand	Summarize threads; detect intent, sentiment and urgency.
Prioritize	Surface replies, waiting threads and approaching deadlines.
Extract	Pull names, dates, amounts, requests and action items.
Draft	Create a reply in the approved user style without sending.
Act	Send, label, archive or schedule only when permission and approval allow it.
Background mode	Use events/webhooks and workers to classify new mail and notify or prepare drafts.

Reply decision examples

Reply type	Default platform response
Interested	Stop automated sequence and create seller handoff.
Question	Use approved knowledge to draft/answer; escalate if uncertain.
Objection	Prepare approved response or route to seller.
Not interested	Stop sequence and record disposition.
Unsubscribe	Immediately suppress future outreach.
Wrong person	Identify or request the correct contact when policy allows.
No reply	Lifecycle agent evaluates whether a follow-up is eligible.

OAuth tokens remain on the backend, email content is treated as untrusted input, and sending is enforced as a backend approval-controlled action.

9. Lead Lifecycle / Next Action Agent

The decision layer that connects all other agents

The Lifecycle Agent answers three questions for every lead: where is the lead now, what just changed, and what should happen next? It maintains one current stage, interprets events, chooses the next action and applies guardrails before action dispatch.

LEAD LIFECYCLE — EVENT TO NEXT BEST ACTION

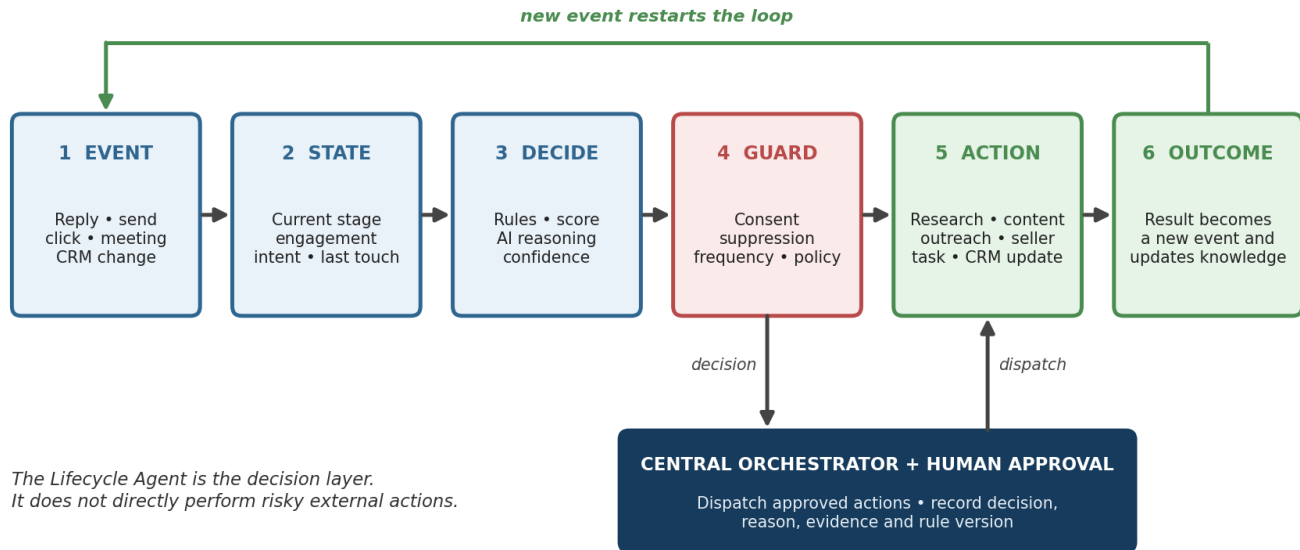


Figure 2. Event-driven lifecycle decision loop.

Decision input	Examples
Context	Lead identity, company, research, campaign and knowledge.
Engagement	Replies, opens, clicks, meetings and recent activity.
Time	Time since last touch, scheduled timers, stalled-lead rules.
Decision methods	Playbook rules, scoring, LLM reasoning and confidence.
Guardrails	Do-not-contact, consent, frequency caps, compliance and approval.
Output	Action, channel, timing, owner, reason and evidence.

10. Agent-to-Agent Interaction and Handoffs

How the specialists work together without becoming one large agent

SPECIALIST AGENT INTERACTION MAP

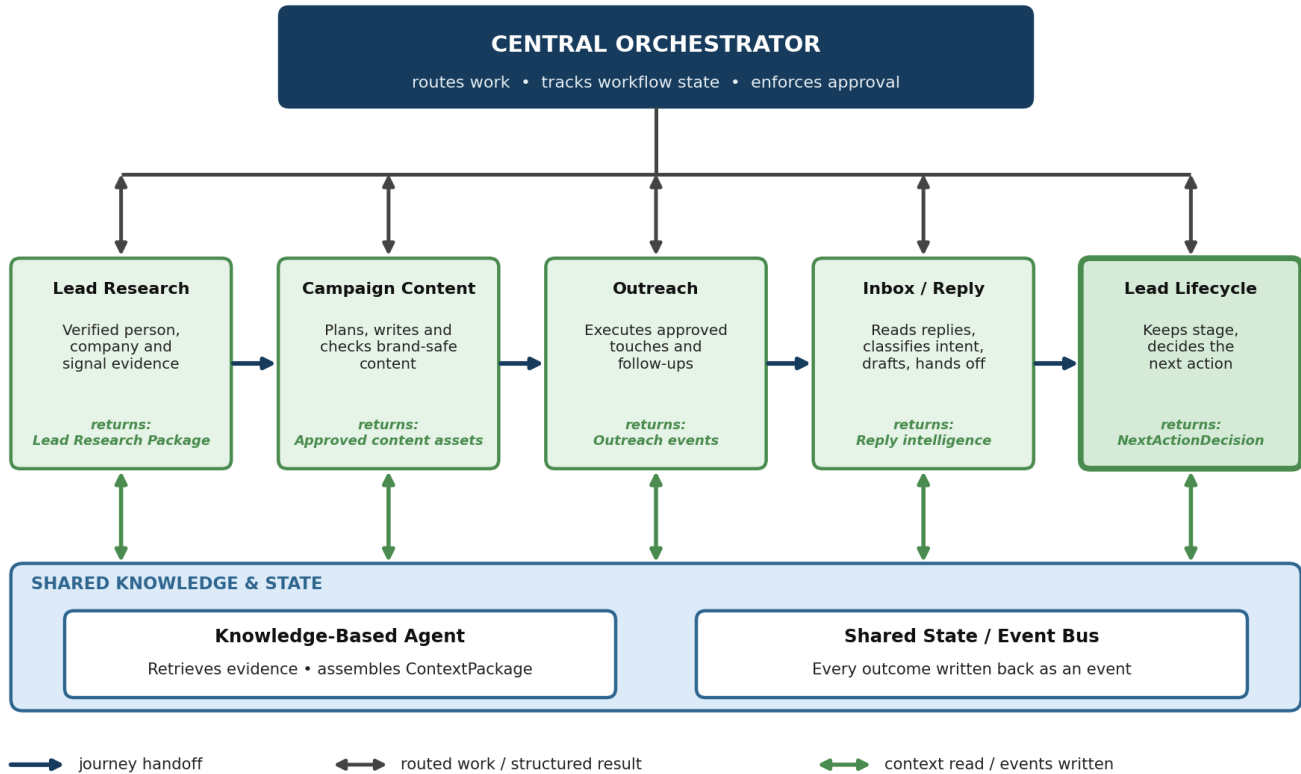


Figure 3. Specialist interaction map. The arrows represent routed work and shared context, while the event layer keeps the system state current.

Typical end-to-end lead journey

1. New lead enters CRM or campaign segment.
2. Lifecycle requests context and research when information is thin.
3. Lead Research creates an evidence-backed research package.
4. Lifecycle evaluates the lead and requests Campaign Content or Outreach work.
5. Campaign Content creates approved messaging; Outreach executes approved touches.
6. Outreach events and inbox replies are written to shared state.
7. Inbox classifies replies and creates a draft or human handoff when needed.
8. Lifecycle reads the new event and chooses the next action.
9. Knowledge-Based Agent updates the shared record with new evidence, interaction history and human corrections.
10. The loop continues until the lead reaches a terminal state such as qualified, converted, lost or suppressed.

11. Data, State and Event Model

The shared data contract that keeps the agents synchronized

A centralized platform needs one reliable representation of the lead and its history. The individual agents can own specialist outputs, but they should not create separate competing lifecycle records.

Core object	Key fields / purpose
Lead	Identity, contact, company, source, lifecycle stage, owner, consent/suppression.
Company	Domain, industry, size, location, technology, business signals.
ResearchPackage	Verified facts, signals, score, evidence, confidence, research timestamp.
Campaign	Goal, audience, channels, offer, schedule, budget, constraints.
ContentAsset	Channel, version, status, evidence, checks, reviewer, publish time.
OutreachEvent	Channel, timestamp, content version, status, provider event ID.
Conversation	Thread, participants, messages, intent, sentiment, open requests, handoff.
NextActionDecision	Action, channel, timing, owner, reason, evidence, confidence, rule version.
KnowledgeItem	Claim, source, timestamp, provenance, permissions, confidence, freshness.
AuditEvent	Actor, tool, action, inputs/outputs metadata, approval and result.

Event-driven pattern

Examples of events include LeadCreated, ResearchCompleted, OutreachSent, EmailReceived, ReplyClassified, MeetingBooked, HumanApproved, ContentPublished and LeadStageChanged. Each event should be idempotent so retries do not create duplicate state changes.

12. Security, Governance and Reliability

Safety is a platform feature, not a prompt instruction

Control	Implementation direction
Least privilege	Give each agent only the tools and scopes required for its job.
Human approval	Require approval for sending, publishing, high-impact decisions and sensitive cases.
Suppression	Check do-not-contact, consent and frequency limits before outreach.
Evidence	Require sources and confidence for important factual claims.
Injection defence	Treat email, web pages and external documents as untrusted input.
Secrets	Keep API keys and OAuth tokens on the backend, encrypted and out of the browser.
Audit	Record important research, tool calls, decisions, approvals and external actions.
Versioning	Version prompts, playbooks, campaign rules and decision policies.
Failure safety	When a source fails or data conflicts, mark uncertainty instead of inventing an answer.
Retention	Define what lead, email and research data is retained and when it is deleted.

Failure handling

- External source unavailable → use available trusted data and clearly mark the gap.
- Retrieval failure → fall back to structured/keyword retrieval where possible; never invent missing context.
- Entity ambiguity → keep the lead unresolved or request clarification.
- Conflicting data → preserve conflict metadata and use source/recency rules rather than silently choosing.
- LLM extraction failure → retry or use deterministic parsing while preserving the raw source.
- Permission failure → exclude restricted information and record the access decision.
- Duplicate event → use event IDs and idempotency keys.

Governance principle

The model may recommend. The platform decides whether the recommendation is permitted, approved and executable.

13. Technical Stack, APIs and Implementation Roadmap

A practical build plan based on the supplied architectures

Layer	Recommended direction
Frontend	React + TypeScript or the existing sales frontend
Backend API	Python + FastAPI
Agent orchestration	LangGraph or application-level workflow orchestration
Validation	Pydantic / typed schemas
Structured data	PostgreSQL
Cache / jobs	Redis + worker system
Semantic retrieval	Production vector database + embeddings
Documents / assets	Object storage + CDN where needed
Email	Gmail API / Microsoft Graph with OAuth 2.0
External actions	Typed integration services with permission checks
Authentication	OAuth2/OIDC + JWT + RBAC
Deployment	Docker-based cloud deployment

Recommended implementation order

1. Build the shared data model, authentication, audit log and central API.
2. Build the Knowledge-Based Agent and ContextPackage contract.
3. Build Lead Research with evidence and scoring.
4. Build Lead Lifecycle as the decision and routing layer.
5. Build Campaign Content with brand grounding and approval.
6. Build Outreach execution with strict suppression and send controls.
7. Build Inbox / Reply Intelligence with OAuth and backend approval for sending.
8. Connect events, analytics and feedback loops across all agents.
9. Add optimization only after quality, safety and state consistency are measurable.

MVP boundary

Start read-heavy and approval-heavy. Research, retrieval, classification and drafting can be automated first. External sending, publishing and other irreversible actions should remain gated until confidence, logging and failure handling are proven.

14. Evaluation, KPIs and Final Architecture Principles

How to know the platform is working

Area	Useful measures
Knowledge	Retrieval precision/recall, groundedness, freshness, entity-resolution accuracy, provenance coverage.
Research	Research completeness, evidence coverage, score consistency, stale-data rate.
Campaign Content	Brand-check pass rate, claim accuracy, approval rate, revision rate, content performance.
Outreach	Qualified lead rate, valid contact rate, bounce rate, reply rate, positive reply rate, meeting rate.
Inbox	Classification accuracy, urgent-email recall, draft acceptance, handoff accuracy, response latency.
Lifecycle	Stage accuracy, next-action acceptance, stalled-lead reduction, policy-block rate.
Platform	Latency, cost per run, tool failure rate, duplicate-event rate, audit coverage.

Final architecture principles

1. One shared state: all agents work from the same current lead and account reality.
2. One shared knowledge layer: evidence is retrieved once and reused across tasks.
3. Specialist agents: keep responsibilities narrow enough to test and govern.
4. Central orchestration: route work and enforce workflow state rather than letting agents call each other freely.
5. Event-driven feedback: every important outcome becomes a new fact for future decisions.
6. Rules + scoring + AI: deterministic policy for clear cases, scoring for priority, AI reasoning for ambiguity.
7. Approval by risk: drafting can be automated; irreversible actions require explicit permission.
8. Evidence before personalization: do not invent prospect facts, claims or business events.
9. Audit everything important: decisions should be explainable after the fact.
10. Measure before optimizing: improve prompts and workflows only after quality and safety are observable.

Final conclusion

The strongest design is not six separate AI bots. It is one governed Sales AI platform with a central orchestrator, a shared knowledge/context service, shared lead state and events, six specialist agents, and controlled human approval. This structure keeps the system modular while allowing every agent to contribute to the same sales journey.

End of report